

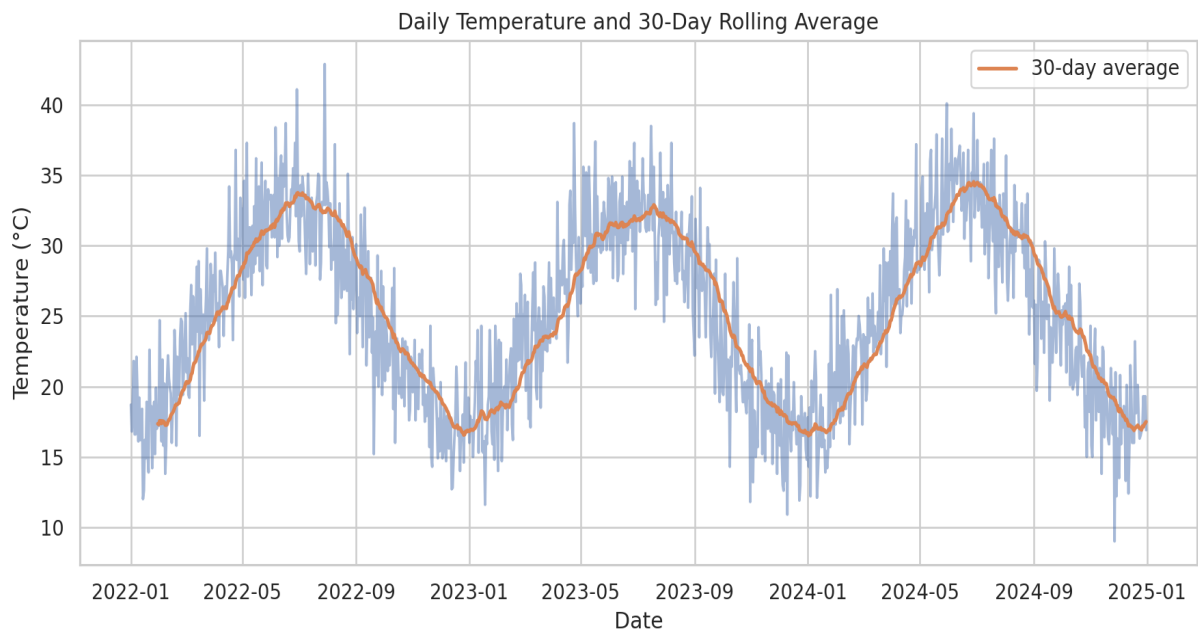
WEATHER DATA ANALYSIS

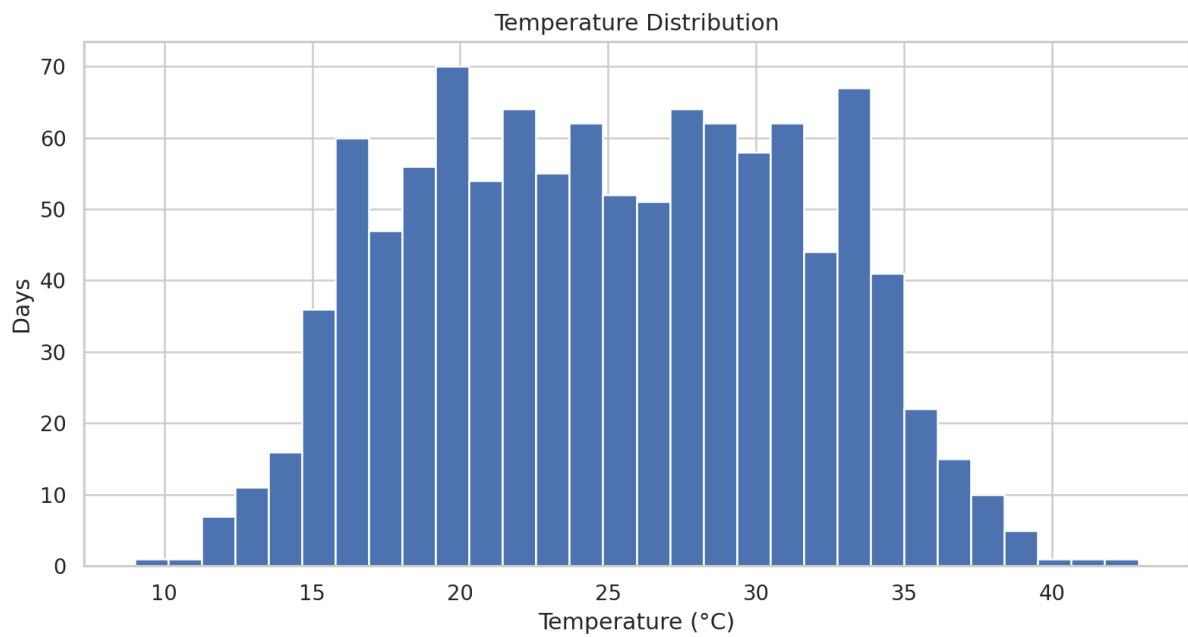
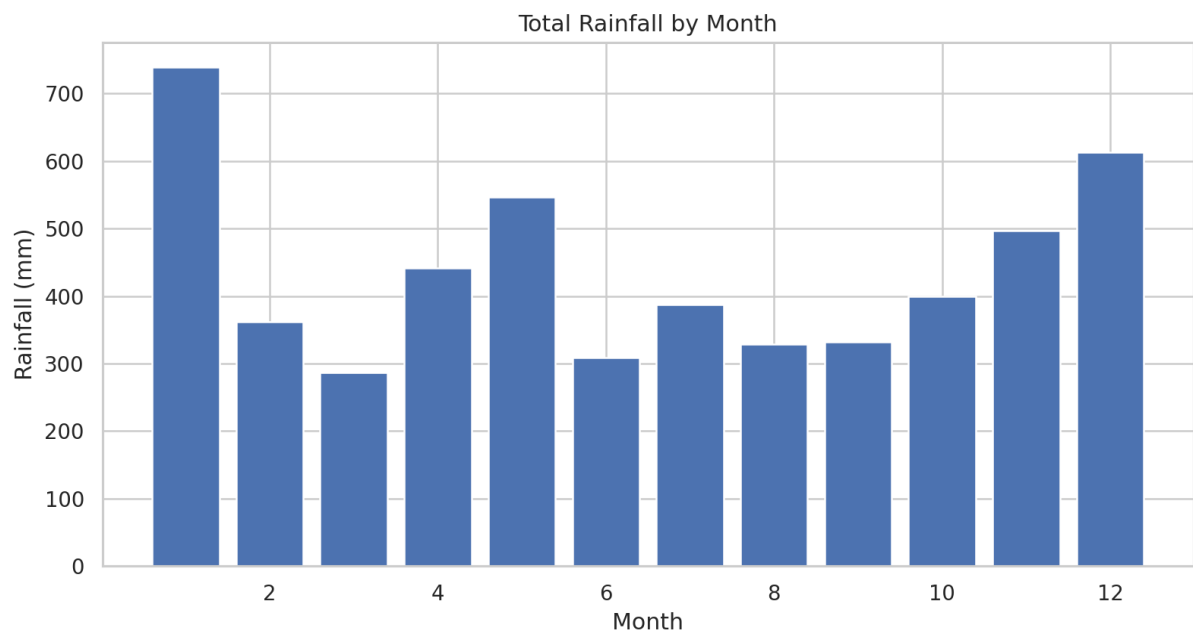
Portfolio Data Analysis Report — Simulated Dataset

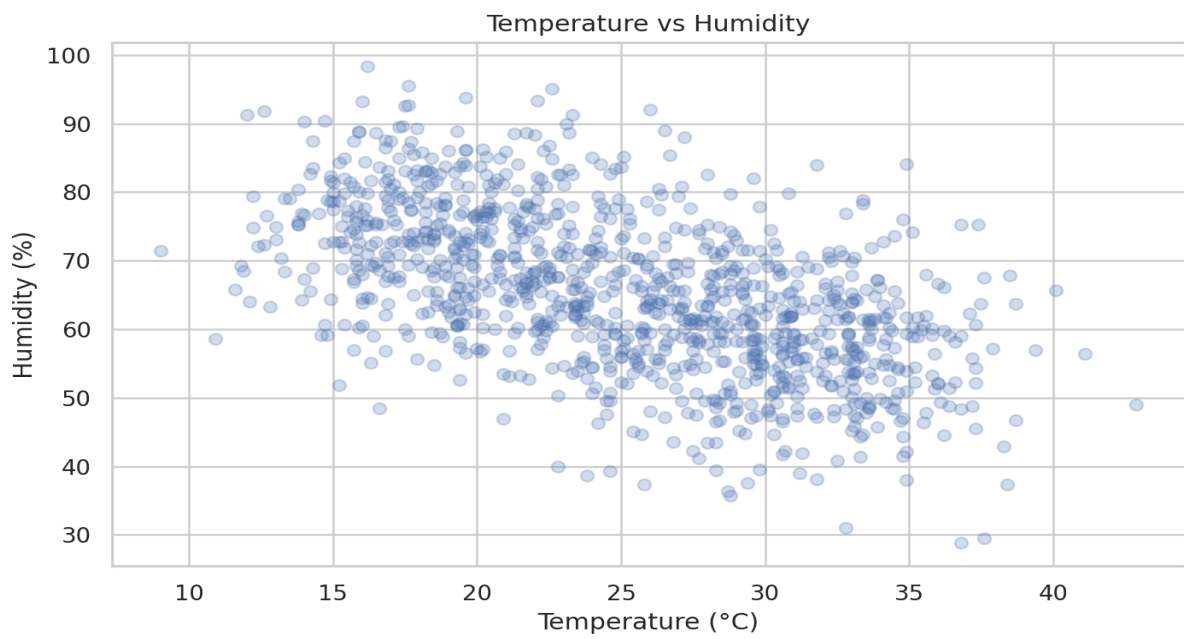
Executive Summary: This project analyzes 1,096 simulated daily observations from 2022-01-01 through 2024-12-31. The analysis focuses on temperature trends, seasonal rainfall, humidity relationships and extreme observations.

Metric	Value
Observations	1,096
Period	2022-01-01 to 2024-12-31
Average temperature	25.10 °C
Total rainfall	5,240.8 mm
Rainy days	279
Temp / humidity correlation	-0.565
Temp / rainfall correlation	-0.080
Extreme-temperature days	41

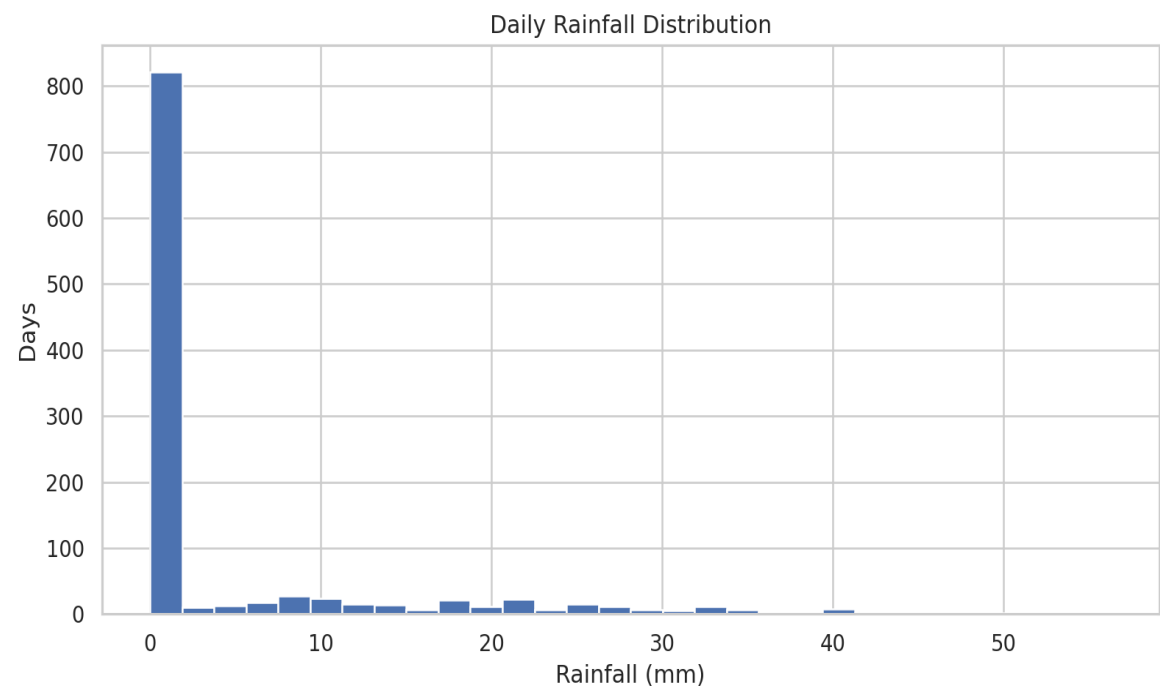
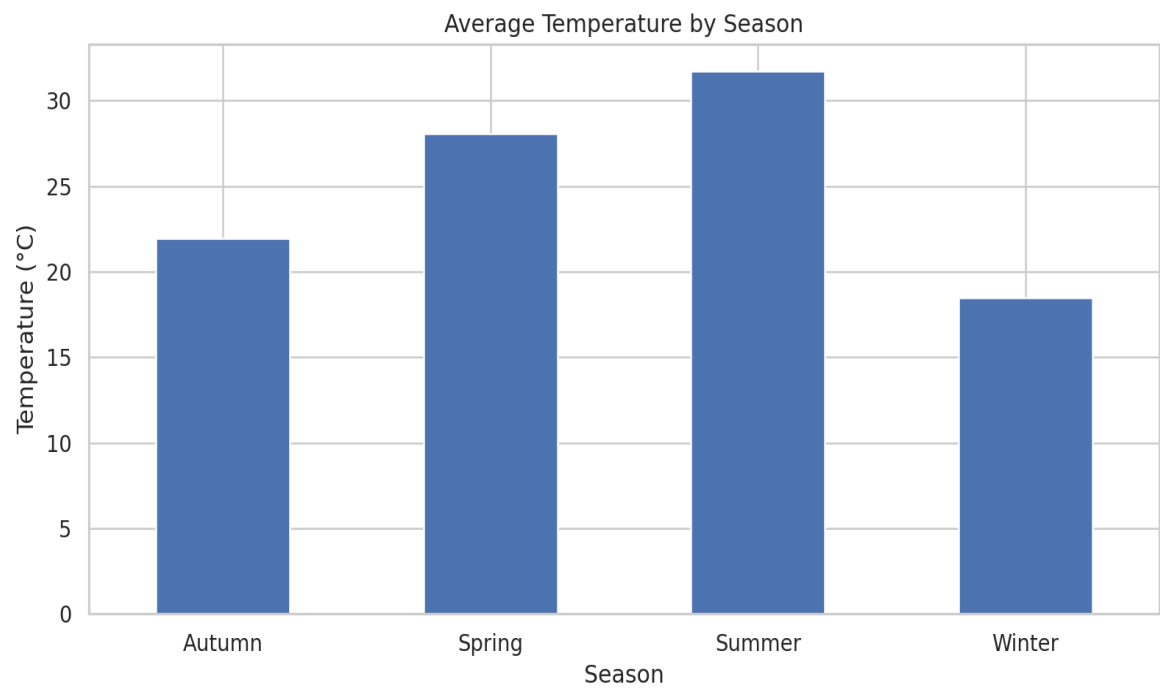
1. Visual Analysis







2. Seasonal Analysis



3. Key Findings

- Temperature exhibits a recurring seasonal pattern.
- The strongest monthly rainfall total occurs in month 1.
- The simulated temperature/humidity correlation is -0.565.
- The simulated temperature/rainfall correlation is -0.080.
- The hottest recorded observation is 42.9 °C on 2022-07-29.
- The wettest recorded observation is 56.3 mm on 2022-03-06.

4. Recommendations

- Use monthly and seasonal summaries for planning.
- Monitor rainfall extremes separately from average rainfall.
- Use rolling averages to distinguish sustained trends from daily noise.
- Use validated, longer-term observational data for real meteorological or climate decisions.

5. Limitation

The dataset is simulated and is intended for demonstrating analysis techniques rather than representing official weather observations.